

Real-branch asymptotic inversion at a singular endpoint

Power–logarithmic expansions with irrational exponents,
convergent reversion, and a Wolfram Language implementation

Research and implementation report
Prepared for Vladimir Reshetnikov

7 September 2026

Abstract

The inverse of $x + x^2(1 + \log x)$ on the positive real axis has logarithmic coefficient blocks, not a Taylor series at zero. The inverse of $x + x^{\sqrt{2}}$ has irrational exponents and does not fit a rational Puiseux grid. We solve both problems within a common constructive theory for

$$f(x) = ax^p \left(1 + \sum_{i=1}^m x^{\delta_i} B_i(\log x) \right), \quad a, p, \delta_i > 0, \quad B_i \in \mathbb{R}[L].$$

An explicit finite differential-operator formula computes every inverse block. The inverse support lies in a positive finitely generated additive semigroup, so every power cutoff requires finitely many multi-indices even when the exponent group is dense. We prove positive-branch existence and uniqueness, the coefficient formula, actual convergence for each finite model sufficiently near zero, a finite-boundary remainder theorem, and transport of differentiated forward remainders. Resonances, logarithmic orders, and nonunit leading powers are treated explicitly.

The accompanying package implements exact algebraic-exponent enumeration, polynomial logarithmic coefficients, complete-block truncation, guarded input precision, and an independent sparse-composition residual check. A separate marker-based reversion function handles general near-identity perturbations without pretending that marker order is automatically asymptotic order. All limitations and verification status are explicit: the mathematical formulas have independent exact and high-precision checks, but native Wolfram-kernel execution was unavailable in this session.

Contents

1	The problem and the answer	3
2	Selecting a real branch before expanding	4
2.1	The first example is globally invertible on the positive axis	4
2.2	The general branch condition	4
3	The algebra of power–logarithmic blocks	5
3.1	Blocks and their actual asymptotic ordering	5
3.2	Irrational exponents do not destroy local finiteness	6
4	Reversion with an auxiliary marker	6
4.1	A derivative formula that never evaluates the singular endpoint	6
4.2	A general exactly invertible dominant core	7
5	The all-order coefficient formula	7
5.1	Powers and the logarithm of the inverse	9
6	Existence, uniqueness, and convergence of the expansion	9
6.1	A triangular formal equation	9
6.2	An analytic proof using finitely many small parameters	9
6.3	Newton iteration and order doubling	10
7	Truncation and rigorous remainder semantics	10
7.1	A finite-boundary theorem	10
7.2	Why input precision has to be transported	12
8	The logarithmic example to all orders	12
8.1	A residual that identifies the correct error scale	13
8.2	An explicit a posteriori error bound	13
9	The irrational-power example and its scalar reduction	14
9.1	A one-variable analytic generating function	14
10	Further examples and reductions	15
10.1	Nonunit leading powers	15
10.2	A resonance and an exact cancellation	15
10.3	Several incommensurable gaps and logarithms	15
10.4	Other finite endpoints and inversion at infinity	15
11	Implementation and its exact contract	16
11.1	Accepted and rejected inputs	16
11.2	Public functions and data	16

11.3 The algorithm in six stages	17
11.4 Independent residual verification	17
11.5 Why not manufacture a rational <code>SeriesData</code> object?	18
12 Verification performed and reproducibility	18
12.1 What was actually executed	18
12.2 Numerical behavior on the positive branch	18
12.3 Running the supplied checks	20
13 Boundaries of the theory implemented here	20
13.1 A small logarithmic correction with zero power gap	20
13.2 A leading logarithm, iterated logarithms, and flat sectors	20
13.3 Symbolic exponents and nonuniform limits	21
13.4 What a larger implementation should add next	21
14 Relationship to the supplied materials and main conclusions	21
A A compact marker recipe	22
B An exact residual bracket lemma	22
C Source and verification manifest	22

1 The problem and the answer

The question supplied with this task, and its public version [1], asks for the positive real inverse as the argument tends to zero. Its two examples are

$$f_1(x) = x + x^2(1 + \log x), \quad f_2(x) = x + x^{\sqrt{2}}, \quad x > 0. \quad (1.1)$$

The archived question reports unsuccessful attempts involving `InverseFunction`, `Series`, `Asymptotic`, and `InverseSeries`. Those are historical observations, not tests of a current Wolfram release performed for this report.

Put $L = \log y$. The first inverse is

$$\begin{aligned} g_1(y) = & y - y^2(1 + L) + y^3(2L^2 + 5L + 3) \\ & - y^4 \left(5L^3 + \frac{41}{2}L^2 + 27L + \frac{23}{2} \right) \\ & + y^5 \left(14L^4 + \frac{241}{3}L^3 + \frac{335}{2}L^2 + 151L + \frac{299}{6} \right) + \mathcal{O}(y^6(1 + |L|)^5). \end{aligned} \quad (1.2)$$

The second inverse is

$$g_2(y) = y - y^{\sqrt{2}} + \sqrt{2}y^{2\sqrt{2}-1} - \frac{6-\sqrt{2}}{2}y^{3\sqrt{2}-2} + \mathcal{O}(y^{4\sqrt{2}-3}). \quad (1.3)$$

Every limit and remainder in these displays is for $y \rightarrow 0^+$. Neither formula requires selecting a complex branch of an unspecified `InverseFunction` object.

Why logarithms must remain in the remainder. After retaining only $y - y^2(1 + \log y)$, the error is

$$y^3(2 \log^2 y + 5 \log y + 3) + \mathcal{O}(y^4(1 + |\log y|)^3). \quad (1.4)$$

It is therefore *not* $\mathcal{O}(y^3)$ in the usual analytic meaning of big-O. A valid bound is $\mathcal{O}(y^3(1 + |\log y|)^2)$. Logarithms may occur inside Wolfram's series coefficients [7, 8]; this does not license dropping them from an analytic remainder bound.

Minimal package use. After extracting the archive, load the file with an appropriate local path:

```
Get["RealInverseAsymptotics/Kernel/RealInverseAsymptotics.wl"];
Clear[x, y];
r = RealInverseAsymptotic[
  x + x^2 (1 + Log[x]), {x, 0}, {y, 6}];
Normal[r]
r["Remainder"]
InverseResidual[r]["ZeroBelowCutoff"]
```

The cutoff is *exclusive*: `{y,6}` retains every complete logarithmic block with power smaller than 6. The predicted outputs are (1.2) without its remainder, `PowerLogRemainder[y,6,5]`, and `True`. These are mathematical expectations encoded in the regression suite, not a claim of a native-kernel run in this session. Section 12 records what was run.

2 Selecting a real branch before expanding

2.1 The first example is globally invertible on the positive axis

Proposition 2.1. *The map $f_1 : (0, \infty) \rightarrow (0, \infty)$ is a strictly increasing bijection. Its inverse satisfies $g_1(y)/y \rightarrow 1$ as $y \rightarrow 0^+$.*

Proof. Differentiation gives

$$f_1'(x) = 1 + 3x + 2x \log x.$$

The function $3x + 2x \log x$ has derivative $5 + 2 \log x$, so its global minimum occurs at $x = \exp(-5/2)$ and is $-2 \exp(-5/2)$. Consequently

$$f_1'(x) \geq m_0 := 1 - 2 \exp(-5/2) > 0 \quad (x > 0). \quad (2.1)$$

Also $f_1(x) \rightarrow 0$ at zero and $f_1(x) \rightarrow \infty$ at infinity. Strict monotonicity and the intermediate value theorem prove the bijection. Finally $f_1(x)/x = 1 + x(1 + \log x) \rightarrow 1$, and substitution $x = g_1(y)$ proves the inverse equivalence. $\square$

The continuous extension has $f_1(0) = 0$ and a right derivative equal to one, but

$$f_1''(x) = 5 + 2 \log x \rightarrow -\infty.$$

Likewise $g_1''(y) = -f_1''(g_1(y))/f_1'(g_1(y))^3 \rightarrow +\infty$. Thus there is no ordinary second-order Taylor expansion of the smooth-at-zero kind that an analytic inverse theorem would require. A nonzero limiting first derivative alone is not analyticity.

The constant in the historical complex output has the form

$$c = \exp(-1 + W_{-1}(-e)).$$

The nonzero zeros of the complex continuation formally satisfy $1 + x(1 + \log x) = 0$, which becomes $te^t = -e$ after $t = 1 + \log x$. A local expansion about such a nonzero complex preimage is a different problem from the real branch characterized by $g_1(y) \rightarrow 0^+$. Reality of some symbols is not a substitute for specifying the limiting preimage and the domain of the inverse. The package never asks `InverseFunction` to choose that branch; it constructs the positive local branch directly. The documented role of `InverseFunction` and the series-reversion role of `InverseSeries` are distinct [10, 9].

2.2 The general branch condition

Our central model is

$$f(x) = ax^p \left(1 + \sum_{i=1}^m x^{\delta_i} B_i(\log x) \right), \quad a > 0, \quad p > 0, \quad \delta_i > 0, \quad (2.2)$$

where each B_i is a real polynomial and all parameters are fixed. The empty sum, $m = 0$, is permitted. Equal δ_i should first be combined; zero blocks may be deleted.

Lemma 2.2 (Positive local branch). *For (2.2), there is an $\varepsilon > 0$ such that f is positive and strictly increasing on $(0, \varepsilon)$. Its inverse g on the corresponding image satisfies*

$$g(y) \sim z, \quad z = (y/a)^{1/p} > 0. \quad (2.3)$$

This characterizes the branch used throughout the report.

Proof. For every $\delta > 0$ and fixed integer $d \geq 0$, $x^\delta(1 + |\log x|)^d \rightarrow 0$. This follows by putting $x = e^{-t}$ and comparing $e^{-\delta t}(1 + t)^d$ with zero as $t \rightarrow \infty$. Differentiating each correction in (2.2) shows

$$f(x) \sim ax^p, \quad f'(x) \sim apx^{p-1} > 0.$$

Positivity and monotonicity follow on a sufficiently small interval. The inverse exists there and tends to zero. Substituting it into the first equivalence gives $y \sim ag(y)^p$, hence (2.3). $\square$

This lemma is local: a general finite model can turn around away from the endpoint. Global invertibility is established separately for the two original examples, not presumed for every package input. Real powers are always taken on positive arguments, using $x^r = \exp(r \log x)$. This restriction is essential; Wolfram’s general complex **Power** uses principal values [11]. The implementation does not apply **PowerExpand** to unknown-sign expressions.

3 The algebra of power–logarithmic blocks

3.1 Blocks and their actual asymptotic ordering

Let $L = \log z$ and $\mathcal{D} = d/dL$. A block is $z^\alpha P(L)$, with P a polynomial. If $\alpha < \beta$ and P, Q are nonzero polynomials,

$$z^\beta Q(\log z) = o(z^\alpha P(\log z)). \quad (3.1)$$

Indeed, the ratio has a positive power of z multiplied by a rational function of $\log z$, and the exponential comparison just used applies. Within one block, larger logarithmic degree means larger magnitude. A fully ordered monomial scale therefore orders the power increasingly and, at equal power, the logarithmic degree decreasingly.

The software retains *whole polynomial blocks*. This avoids treating L as constant during differentiation or silently discarding a lower logarithmic term at a retained power. It also gives an unambiguous meaning to a cutoff. A request for “four terms” would otherwise be ambiguous: there may be four monomials inside just two power blocks.

Lemma 3.1 (Differential block identity). *For every nonnegative integer r ,*

$$\frac{d^r}{dz^r} [z^\alpha P(\log z)] = z^{\alpha-r} \prod_{j=0}^{r-1} (\mathcal{D} + \alpha - j) P(L). \quad (3.2)$$

The empty product, for $r = 0$, is the identity operator.

Proof. One differentiation gives $z^{\alpha-1}(\alpha P + P')$. Repeated application proves the result by induction. The factors commute because each is a constant plus $\mathcal{D}$. $\square$

This identity is the computational bridge from a derivative formula to finite polynomial arithmetic. A derivative of $P(\log z)$ is never omitted. For instance,

$$\frac{1}{2} \frac{d}{dy} \{y^4(1 + \log y)^2\} = y^3 \{2(1 + \log y)^2 + (1 + \log y)\},$$

which is exactly the third block of (1.2).

3.2 Irrational exponents do not destroy local finiteness

For the positive gaps in (2.2), set

$$S = \{n_1\delta_1 + \cdots + n_m\delta_m : n_i \in \mathbb{N}_0\}. \quad (3.3)$$

If $\sum n_i\delta_i < H$, then $n_i < H/\delta_i$ for every i . Consequently there are only finitely many admissible multi-indices below any finite H , and only finitely many support values there. The additive group generated by two incommensurable gaps may be dense, but it allows negative integer coefficients. The positive semigroup (3.3) does not have that defect.

We can therefore work with formal sums

$$U(z) = \sum_{\sigma \in S \setminus \{0\}} z^\sigma C_\sigma(L), \quad C_\sigma \in \mathbb{R}[L],$$

in a completion with respect to increasing power order. Products, and the formal series $\log(1+U)$ and $(1+U)^r$, are defined coefficientwise: at every fixed cutoff only finitely many products contribute. The valuation $\text{val } U$ is the least nonzero power in its support; logarithmic degree is tracked separately. This restricted, constructive setting is enough here. General transseries and their compositional inversion live in a substantially larger framework [3, 4].

Resonances. Different multi-indices can have the same weight. If $\delta_2 = 2\delta_1$, then $(2, 0)$ and $(0, 1)$ coincide in weight. Their polynomial coefficients must be added, not counted as different asymptotic terms. Cancellation can remove an entire block. Exact algebraic canonicalization in the package is used precisely for this reason; floating-point ordering or tolerance-based equality would be an unsound replacement.

4 Reversion with an auxiliary marker

4.1 A derivative formula that never evaluates the singular endpoint

The underlying identity is a standard form of Lagrange–Bürmann inversion [2]. We give its proof to make clear what is analytic, what is formal, and why logarithms at zero are not an obstruction.

Theorem 4.1 (Marker reversion). *Fix $q > 0$. Suppose h and Φ are analytic in a complex neighborhood of q . Let $u(q, \eta)$ be the solution of*

$$u + \eta h(u) = q, \quad u(q, 0) = q.$$

Then, for sufficiently small $|\eta|$,

$$\Phi(u(q, \eta)) = \Phi(q) + \sum_{N \geq 1} \frac{(-\eta)^N}{N!} \frac{d^{N-1}}{dq^{N-1}} [\Phi'(q)h(q)^N]. \quad (4.1)$$

It is also an identity of formal power series in the marker.

Proof. Put $u = q + s$, $\phi(s) = -h(q + s)$, and $\Psi(s) = \Phi(q + s)$. Then $s = \eta\phi(s)$. Initially assume $\phi(0) \neq 0$. Changing variable $\eta = s/\phi(s)$ in the residue for the coefficient of η^N gives

$$[\eta^N]\Psi(s(\eta)) = \text{Res}_{s=0} \Psi(s) \left\{ \frac{\phi(s)^N}{s^{N+1}} - \frac{\phi'(s)\phi(s)^{N-1}}{s^N} \right\} ds$$

$$= \frac{1}{N} \operatorname{Res}_{s=0} \frac{\Psi'(s)\phi(s)^N}{s^N} ds = \frac{1}{N} [s^{N-1}] \Psi'(s)\phi(s)^N.$$

The middle equality follows because the residue of $d(\Psi(s)\phi(s)^N/s^N)/ds$ is zero. Replacing the coefficient by a Taylor derivative gives (4.1). Each coefficient is a polynomial in finitely many derivatives of h and Φ , so the identity extends to $\phi(0) = 0$ as well. Analytic existence and convergence near $\eta = 0$ follow independently from the analytic implicit function theorem, since the derivative with respect to u at $(q, 0)$ is one. $\square$

For $\Phi(q) = q$, this becomes the particularly simple rule

$$g(y) = y + \sum_{N \geq 1} \frac{(-1)^N}{N!} \frac{d^{N-1}}{dy^{N-1}} h(y)^N, \quad f(x) = x + h(x). \quad (4.2)$$

In (4.2), setting the marker to one needs justification. Formal positive grading supplies a coefficientwise justification; an analytic smallness argument is still required to identify the sum with an actual function. Section 6 supplies both for our finite models. The distinction is not cosmetic: a marker Taylor series can exist without being a useful asymptotic expansion at the endpoint of interest.

At each fixed positive q , functions such as q^α and $\log q$ are analytic in a small complex disk avoiding zero and the negative axis. No derivative at $q = 0$ occurs in the theorem. This is the essential reason that the derivative form works for the question while an ordinary Taylor inverse at zero is the wrong object.

4.2 A general exactly invertible dominant core

A useful generalization explains the connection with the supplied transseries materials. Suppose

$$F(x) = H(x) + R(x), \quad C = H^{-1}$$

on a specified local branch. Put $t = H(x)$ and $h(t) = R(C(t))$. Then $t + \eta h(t) = y$ and (4.1) gives

$$F_\eta^{-1}(y) = C(y) + \sum_{N \geq 1} \frac{(-\eta)^N}{N!} \frac{d^{N-1}}{dy^{N-1}} [C'(y)R(C(y))^N]. \quad (4.3)$$

This is a change of coordinate, not a new Lagrange inversion theorem. The companion volume supplied in the repository uses precisely this kind of phase-coordinate and scaled reversion, and explicitly distinguishes formal marker identities from convergence at the physical marker [6]. Here the core is $H(x) = ax^p$, and the positive power gaps provide a finite, directly implementable grading.

5 The all-order coefficient formula

For a multi-index $\mathbf{n} = (n_1, \dots, n_m) \in \mathbb{N}_0^m$, write

$$N = |\mathbf{n}| = \sum_i n_i, \quad \mathbf{n}! = \prod_i n_i!, \quad \sigma(\mathbf{n}) = \sum_i n_i \delta_i, \quad B^\mathbf{n}(L) = \prod_i B_i(L)^{n_i}. \quad (5.1)$$

For $N \geq 1$, define

$$C_\mathbf{n}(L) = \frac{(-1)^N}{p^N \mathbf{n}!} \prod_{j=1}^{N-1} (\mathcal{D} + 1 + \sigma(\mathbf{n}) + pj) B^\mathbf{n}(L). \quad (5.2)$$

Again the operator product is empty when $N = 1$.

Theorem 5.1 (Constructive inverse formula). *For the positive branch of (2.2), with $z = (y/a)^{1/p}$ and $L = \log z$, the inverse has the expansion*

$$g(y) = z + \sum_{\mathbf{n} \neq 0} z^{1+\sigma(\mathbf{n})} C_{\mathbf{n}}(L). \quad (5.3)$$

Equal weights are combined by summing their $C_{\mathbf{n}}$. The formula is a well-defined formal identity. For each fixed finite model, it also converges absolutely for sufficiently small positive y in the sense established in Section 6. In particular, every finite cutoff is a genuine asymptotic expansion of the specified real inverse.

Proof of the coefficient identity. Set $t = x^p$ and $q = y/a$. Equation (2.2) becomes

$$q = t + \sum_i t^{1+\delta_i/p} B_i \left(\frac{\log t}{p} \right).$$

Insert a separate marker for each summand, apply (4.1) to $\Phi(t) = t^{1/p}$, and extract the indicated multinomial coefficient. The coefficient belonging to $\mathbf{n}$ is

$$\frac{(-1)^N}{\mathbf{n}!} \frac{d^{N-1}}{dq^{N-1}} \left[\frac{1}{q} q^{N-1+(1+\sigma)/p} B^{\mathbf{n}} \left(\frac{\log q}{p} \right) \right]. \quad (5.4)$$

Applying (3.2), with $d/d(\log q) = \mathcal{D}/p$, leaves the power $q^{(1+\sigma)/p} = z^{1+\sigma}$. Each of the $N-1$ derivative factors contributes a factor $1/p$, in addition to the displayed $1/p$. The resulting factors are exactly $\mathcal{D} + 1 + \sigma + p, \dots, \mathcal{D} + 1 + \sigma + p(N-1)$, proving (5.2). Local finiteness of (3.3) makes every coefficient and every cutoff finite, so the marker substitution is formally legitimate. Analytic convergence is proved below without presuming it from this formal calculation. $\square$

Corollary 5.2 (Degree and support bounds). *If $d_i = \deg B_i$, then*

$$\deg C_{\mathbf{n}} \leq D(\mathbf{n}) := \sum_i n_i d_i. \quad (5.5)$$

Every inverse power belongs to $\{(1+\sigma)/p : \sigma \in S\}$. Cancellations can remove powers but cannot create a power outside this set.

Proof. The initial product has degree $D(\mathbf{n})$; differentiation and addition of a scalar multiple do not increase polynomial degree. The support statement is immediate from (5.3). $\square$

First and second corrections. For a single block $x^\delta B(L)$, the first two corrections are

$$g(y) = z - \frac{z^{1+\delta}}{p} B(L) + \frac{z^{1+2\delta}}{2p^2} \left\{ (1+2\delta+p) B(L)^2 + 2B(L)B'(L) \right\} + \dots \quad (5.6)$$

For two different blocks, the multi-index $e_i + e_j$ gives

$$\frac{z^{1+\delta_i+\delta_j}}{p^2} (\mathcal{D} + 1 + \delta_i + \delta_j + p) (B_i B_j). \quad (5.7)$$

These cross terms are why simply adding separately computed inverse corrections is incorrect.

5.1 Powers and the logarithm of the inverse

The same proof with $\Phi(t) = t^{r/p}$, for any fixed real r , yields

$$g(y)^r = z^r + \sum_{n \neq 0} \frac{r(-1)^N z^{r+\sigma}}{p^N n!} \prod_{j=1}^{N-1} (\mathcal{D} + r + \sigma + pj) B^n(L). \quad (5.8)$$

For $r = 0$ the expression is identically one. Differentiating with respect to r at zero gives

$$\log g(y) = \log z + \sum_{n \neq 0} \frac{(-1)^N z^\sigma}{p^N n!} \prod_{j=1}^{N-1} (\mathcal{D} + \sigma + pj) B^n(L). \quad (5.9)$$

These are useful theoretical extensions for observables of an inverse. Version 1.0 implements g itself and the marker formula; it does not export a separate observable-power API. Ordinary symbolic operations on a finite `Normal` result must not be confused with automatically propagated asymptotic error bounds.

6 Existence, uniqueness, and convergence of the expansion

6.1 A triangular formal equation

Write $g(y) = z(1 + U)$ and define

$$E(U, z, L) = (1 + U)^p - 1 + \sum_i z^{\delta_i} (1 + U)^{p+\delta_i} B_i(L + \log(1 + U)). \quad (6.1)$$

The inverse equation is $E = 0$, and U has strictly positive valuation. At any new support value σ , the unknown coefficient of z^σ enters as pC_σ plus an expression determined by lower weights. Hence every coefficient is determined by one division by p . This proves formal existence recursively and formal uniqueness. It also provides a second coefficient algorithm independent of Lagrange inversion.

More explicitly, if U and V are two solutions with the same leading constant, then

$$E(U) - E(V) = (U - V)(p + \text{positive-order terms}).$$

The second factor is a unit, so $U = V$. The right inverse is also a left inverse: composing it with f gives the solution asymptotic to x of $f(X) = f(x)$, and the same nonzero-leading-factor argument forces $X = x$.

6.2 An analytic proof using finitely many small parameters

A formal series alone does not establish an asymptotic statement. Here a particularly concrete analytic realization is available.

Theorem 6.1 (Convergence for a finite power-log model). *For fixed real parameters satisfying (2.2), the sum (5.3), including its logarithmic polynomials, is absolutely convergent for all sufficiently small positive y and equals the positive inverse selected in Section 2.*

Proof. Write $w = 1 + U$ and $B_i(v) = \sum_{r=0}^{d_i} b_{ir} v^r$. Expansion in the logarithmic argument gives

$$B_i(L + \log w) = \sum_{j=0}^{d_i} L^j A_{ij}(\log w), \quad A_{ij}(v) = \sum_{r=j}^{d_i} b_{ir} \binom{r}{j} v^{r-j}.$$

Introduce independent complex parameters T_{ij} and the equation

$$\mathcal{E}(w, T) = w^p - 1 + \sum_{i,j} T_{ij} w^{p+\delta_i} A_{ij}(\log w) = 0. \quad (6.2)$$

On a fixed complex disk about $w = 1$ of radius less than one, select the analytic logarithm with $\log 1 = 0$. All powers and logarithms in this formula are analytic there. At $(w, T) = (1, 0)$, $\mathcal{E} = 0$ and $\partial_w \mathcal{E} = p \neq 0$. The analytic implicit function theorem gives an analytic function $w = W(T)$ on a polydisc about zero, with $W(0) = 1$.

Now specialize

$$T_{ij} = z^{\delta_i} (\log z)^j. \quad (6.3)$$

Every component tends to zero as $z \rightarrow 0^+$, so the convergent Taylor series of W applies. Its value is real and close to one, hence positive, because all coefficients and the physical parameters are real. It solves the original inverse equation and therefore agrees with the unique local positive inverse.

The Taylor series is absolutely convergent in a smaller polydisc. Group its monomials by the total number n_i of parameters with first index i . For each such multi-index there are only finitely many distributions among the logarithmic labels j . Absolute convergence permits regrouping. Marker coefficient extraction, or formal uniqueness of (6.1), identifies the grouped coefficient with (5.2). This proves the claimed convergence and equality. $\square$

The theorem does not give an ordinary radius of convergence in y for a Taylor series: logarithms and irrational powers are still present. It gives a convergent analytic function of finitely many small composite parameters. Nor does it give a uniform neighborhood as coefficients or gaps vary. Fixed-parameter asymptotics is the assertion proved here.

6.3 Newton iteration and order doubling

A useful alternative implementation would iterate

$$U_{k+1} = U_k - \frac{E(U_k)}{\partial_U E(U_k)}, \quad U_0 = 0, \quad (6.4)$$

with all operations truncated in the positive-weight algebra. Since $\partial_U E = p +$ positive-order terms, division is by a unit. If U_* is the formal solution and $\text{val}(U_k - U_*) \geq s > 0$, a Taylor expansion in the displacement shows

$$\text{val}(U_{k+1} - U_*) \geq 2s.$$

Indeed, the linear displacement is canceled by the Newton quotient, while all coefficients of the remaining quadratic and higher powers have nonnegative valuation. Initially $s \geq \min_i \delta_i$. This is a formal order-doubling statement, not a floating-point convergence claim. Version 1.0 uses the explicit multi-index formula instead: it is easy to audit, exposes the support, and provides a convenient boundary for the remainder. Its residual checker implements the sparse composition needed for an eventual Newton engine, but is not itself a Newton solver.

7 Truncation and rigorous remainder semantics

7.1 A finite-boundary theorem

Suppose the requested cutoff in the target variable is $q > 1/p$. Define

$$H = pq - 1, \quad I_H = \{\mathbf{n} \in \mathbb{N}_0^m : \sigma(\mathbf{n}) < H\}. \quad (7.1)$$

This is a finite downward-closed set. Its one-step outer boundary is

$$\partial I_H = \{\mathbf{n} + e_i : \mathbf{n} \in I_H, 1 \leq i \leq m\} \setminus I_H. \quad (7.2)$$

The boundary need not be a minimal generating set. It is nevertheless finite and every excluded multi-index dominates at least one boundary index componentwise: follow a path from zero to it, increasing one coordinate at a time, and take the first point leaving I_H .

If $m > 0$, put

$$\sigma_* = \min_{\nu \in \partial I_H} \sigma(\nu), \quad k_* = \max_{\substack{\nu \in \partial I_H \\ \sigma(\nu) = \sigma_*}} D(\nu), \quad \beta_* = (1 + \sigma_*)/p. \quad (7.3)$$

Let G_q be the sum in (5.3) over I_H .

Theorem 7.1 (Remainder from a finite boundary). *For the finite exact model,*

$$g(y) - G_q(y) = \mathcal{O}\left(y^{\beta_*}(1 + |\log y|)^{k_*}\right). \quad (7.4)$$

In particular $\beta_ \geq q$. If $m = 0$, the inverse is exactly $(y/a)^{1/p}$ and the remainder is zero.*

Proof. Use the analytic parameter representation from the preceding section. By a Cauchy estimate on a smaller polydisc, the absolute values of grouped Taylor coefficients admit a convergent geometric majorant in the variables

$$u_i = Cz^{\delta_i}(1 + |L|)^{d_i},$$

for some fixed C . The finite number of logarithmic labels per block is absorbed by decreasing the polydisc radius. The contribution of all terms whose grouped multi-index is outside I_H is therefore bounded by a constant times

$$\sum_{\nu \in \partial I_H} u^\nu \prod_i (1 - u_i)^{-1}$$

for sufficiently small z . This can overcount terms, which is harmless for an absolute bound. The geometric factors remain bounded because $u_i \rightarrow 0$. Undoing the definition of u_i gives an error in $w = g/z$ bounded by a finite sum of $z^{\sigma(\nu)}(1 + |L|)^{D(\nu)}$.

Among a finite set of such bounds, the smallest power dominates; at that power the largest logarithmic degree dominates. Equation (3.1) absorbs every larger power, regardless of its finite logarithmic degree. Multiplication by z and the fixed substitution $z = (y/a)^{1/p}$ prove (7.4). $\square$

What this bound does and does not say. β_* is the first *possible* omitted support value. It is not necessarily the first nonzero omitted coefficient, because resonances or special parameter values can cancel it. The reported bound remains valid in that case, but may not be sharp. Similarly k_* is a safe degree bound, not a guarantee that its leading logarithmic coefficient is nonzero. No numerical constant or explicit interval $0 < y < y_0$ is computed by this theorem or by the package.

The inert object `PowerLogRemainder[y,b,k]` means precisely $\mathcal{O}(y^b(1 + |\log y|)^k)$ as $y \rightarrow 0^+$. It is not an ordinary expression to which power-series arithmetic should be applied. In particular, taking `Normal` discards error metadata; subsequent symbolic operations on that expression do not automatically propagate the error.

7.2 Why input precision has to be transported

Often a complicated elementary function is represented by a finite forward jet rather than by an exact finite formula. Suppose

$$F(x) = f(x) + R(x), \quad R(x) = \mathcal{O}(x^\rho M(x)^k), \quad R'(x) = \mathcal{O}(x^{\rho-1} M(x)^k), \quad (7.5)$$

where $M(x) = 1 + |\log x|$, $\rho > p$, and $k \geq 0$ is an integer.

Theorem 7.2 (Forward-to-inverse precision transport). *Under (7.5), F has a positive local inverse and*

$$F^{-1}(y) - f^{-1}(y) = \mathcal{O}\left(y^{(\rho-p+1)/p}(1 + |\log y|)^k\right). \quad (7.6)$$

Proof. The bounds imply $F(x) \sim ax^p$ and $F'(x) \sim apx^{p-1}$, so its positive inverse exists and is asymptotic to $z = (y/a)^{1/p}$. Let $u = F^{-1}(y)$ and $v = f^{-1}(y)$; then both are asymptotic to z . The residual at v is $F(v) - y = R(v) = \mathcal{O}(z^\rho M(z)^k)$. On the interval joining u and v , the derivative of F is asymptotic, uniformly along this shrinking comparable interval, to apz^{p-1} . The mean value theorem therefore gives $u - v = \mathcal{O}(z^{\rho-p+1} M(z)^k)$. Converting from z to y gives the result. $\square$

The derivative condition is a hypothesis, not a consequence of the value bound. For example, $R(x) = x^2 \sin(x^{-2})$ is $\mathcal{O}(x^2)$, but $R'(x) = 2x \sin(x^{-2}) - 2x^{-1} \cos(x^{-2})$ can destroy the monotonicity of $x + R(x)$. An independently proved monotonicity hypothesis can sometimes replace derivative control for value-error transport; the package uses the simple differentiated-remainder contract to avoid silently assuming it.

The option `InputRemainder -> {rho,k}` records these hypotheses. The implementation requires ρ to exceed every supplied forward power, and rejects a requested cutoff larger than

$$q_{\text{input}} = (\rho - p + 1)/p. \quad (7.7)$$

This is a deliberately conservative interface. Without more detailed logarithmic precision at the boundary, coefficients at or beyond that power are not justified. The final remainder is the larger of the finite model's truncation error and the transported input error: compare powers first, and take the larger logarithmic degree when powers agree.

A correct elementary-function workflow. For $F(x) = e^x - 1 + x^{\sqrt{2}}$, take the Taylor polynomial of the analytic part through x^4 and retain the exact irrational-power term:

```
jet = Normal[Series[Exp[x] - 1, {x, 0, 4}]] + x^Sqrt[2];
r = RealInverseAsymptotic[jet, {x, 0}, {y, 5},
  InputRemainder -> {5, 0}];
```

Taylor's theorem supplies both bounds in (7.5). Dropping an 0 term with `Normal` and omitting `InputRemainder` would instead ask the package to invert the finite polynomial model *exactly*. That is a different mathematical input.

8 The logarithmic example to all orders

Here $a = p = \delta_1 = 1$ and $B_1(L) = 1 + L$. Formula (5.3) takes the simple form

$$g_1(y) = y + \sum_{n \geq 1} y^{n+1} P_n(\log y), \quad P_n(L) = \frac{(-1)^n}{n!} \prod_{j=n+2}^{2n} (\mathcal{D} + j)(1 + L)^n. \quad (8.1)$$

This also follows immediately from (4.2) by differentiating $y^{2n}(1 + \log y)^n$. The two forms are useful cross-checks of the operator normalization.

Proposition 8.1 (Highest logarithms are Catalan coefficients). P_n has degree exactly n , and

$$[L^n]P_n(L) = (-1)^n \frac{1}{n+1} \binom{2n}{n}. \quad (8.2)$$

Consequently, after $n = N$ in (8.1), the remainder is

$$\mathcal{O}(y^{N+2}(1 + |\log y|)^{N+1}), \quad (8.3)$$

and the displayed power-log order is attained by the next block.

Proof. A derivative lowers polynomial degree. To obtain the coefficient of L^n , select the scalar part in every operator factor. This gives

$$\frac{(-1)^n}{n!} \prod_{j=n+2}^{2n} j = (-1)^n \frac{(2n)!}{n!(n+1)!} \neq 0.$$

The boundary index is $N + 1$, of weight $N + 1$ and logarithmic degree $N + 1$, so the remainder theorem gives (8.3). Its first omitted coefficient has the nonzero leading logarithm just calculated. $\square$

The first four P_n appear in (1.2). Two more are

$$P_5(L) = -42L^5 - \frac{3727}{12}L^4 - \frac{5365}{6}L^3 - 1256L^2 - \frac{5177}{6}L - \frac{2789}{12}, \quad (8.4)$$

$$P_6(L) = 132L^6 + \frac{5981}{5}L^5 + \frac{52937}{12}L^4 + \frac{16979}{2}L^3 + 9007L^2 + \frac{30019}{6}L + \frac{22769}{20}. \quad (8.5)$$

The coefficient table in the archive is generated independently by SymPy. Its highest logarithms agree with (8.2), and direct sparse substitution cancels the normalized residual at every retained power.

8.1 A residual that identifies the correct error scale

Let $G_2(y) = y - y^2(1 + L)$. Substitution and expansion give

$$\frac{f_1(G_2(y))}{y} - 1 = -y^2(2L^2 + 5L + 3) + \mathcal{O}(y^3(1 + |L|)^3). \quad (8.6)$$

Since $f'_1 \rightarrow 1$, the forward residual and the inverse error have the same leading magnitude and opposite signs under this normalization. This gives an independent explanation of the missing third block and the invalidity of a bare $\mathcal{O}(y^3)$ in (1.4).

8.2 An explicit a posteriori error bound

For any positive approximation G and any $y > 0$, the global derivative bound (2.1) implies

$$|G - g_1(y)| \leq \frac{|f_1(G) - y|}{1 - 2e^{-5/2}}. \quad (8.7)$$

This follows directly from the mean value theorem, not from a formal series identity. When the residual is evaluated with rigorous interval arithmetic, it can be made into a rounding-certified numerical enclosure. Ordinary high-precision evaluation is evidence, not a substitute for such an interval computation. The supplied numerical tables use high precision and are explicitly diagnostics rather than interval certificates.

9 The irrational-power example and its scalar reduction

The entire family

$$f_\alpha(x) = x + x^\alpha, \quad \alpha > 1, \quad (9.1)$$

is globally increasing on $(0, \infty)$ because $f'_\alpha(x) = 1 + \alpha x^{\alpha-1} > 1$. Its endpoint limits are zero and infinity. Thus the positive inverse is unambiguous.

Theorem 9.1 (Generalized Fuss–Catalan inverse coefficients). *For fixed $\alpha > 1$,*

$$g_\alpha(y) = y + \sum_{n \geq 1} \frac{(-1)^n}{n} \binom{\alpha n}{n-1} y^{1+n(\alpha-1)}. \quad (9.2)$$

The binomial coefficient here is the finite product

$$\frac{1}{n} \binom{\alpha n}{n-1} = \frac{1}{n!} \prod_{j=0}^{n-2} (\alpha n - j) = \frac{1}{1 + n(\alpha - 1)} \binom{\alpha n}{n}. \quad (9.3)$$

No rationality assumption on α is needed.

Proof. Apply (4.2) to $h(y) = y^\alpha$ and differentiate $y^{\alpha n}$ exactly $n - 1$ times. This gives the finite product in (9.3) and the power in (9.2). The other two forms are elementary rearrangements of the same product. $\square$

For $\alpha = \sqrt{2}$, the first coefficients are -1 , $\sqrt{2}$, and $-(6 - \sqrt{2})/2$, reproducing the requested result. After correction $n = N$, the remainder is

$$\mathcal{O}\left(y^{1+(N+1)(\alpha-1)}\right). \quad (9.4)$$

The package call that retains exactly the leading term and three corrections in this example is

```
r = RealInverseAsymptotic[
  x + x^Sqrt[2], {x, 0}, {y, 4 Sqrt[2] - 3}];
Normal[r]
```

A power cutoff, rather than a count of ordinary integer powers, is the relevant request.

9.1 A one-variable analytic generating function

Put $t = y^{\alpha-1}$ and $g_\alpha(y) = yV(t)$. The defining equation becomes

$$V + tV^\alpha = 1, \quad V(0) = 1. \quad (9.5)$$

It is analytic in (V, t) near $(1, 0)$, so V has an ordinary convergent Taylor series in t . Equation (9.2) is exactly that Taylor series multiplied by y . This simple reduction is excellent for a single gap; with several incommensurable gaps there need not be a single parameter producing an integer power grid.

In fact the radius of convergence of V is

$$R_\alpha = \frac{(\alpha - 1)^{\alpha-1}}{\alpha^\alpha}. \quad (9.6)$$

To verify it directly, let c_n be the positive number in (9.3). Its gamma-function form and Stirling's formula give

$$c_n \sim \frac{\sqrt{\alpha}}{\sqrt{2\pi}(\alpha - 1)^{3/2}} n^{-3/2} R_\alpha^{-n}. \quad (9.7)$$

The root test proves (9.6), and the $n^{-3/2}$ factor also proves absolute convergence at $|t| = R_\alpha$. As a consistency check, the first real critical point of (9.5) is $V = \alpha/(\alpha - 1)$, $t = -R_\alpha$, obtained by setting its derivative with respect to V to zero.

This result is not uniform as $\alpha \downarrow 1$: the gap $\alpha - 1$ tends to zero, the asymptotic hierarchy collapses, and at $\alpha = 1$ the inverse is $y/2$, not $y + o(y)$. The software consequently requires fixed numerical algebraic exponents instead of pretending to order symbolic gaps uniformly in a parameter.

10 Further examples and reductions

10.1 Nonunit leading powers

Consider

$$f(x) = 3x^2\{1 + x(1 + \log x)\}.$$

With $z = \sqrt{y/3}$ and $L = \log z$,

$$g(y) = z - \frac{1}{2}z^2(1 + L) + \frac{1}{8}z^3(5L^2 + 12L + 7) + \mathcal{O}(y^2(1 + |\log y|)^3). \quad (10.1)$$

The first two corrections follow from (5.6) with $p = 2$, $\delta = 1$. Notice both the factor p^{-N} in the coefficient and the change $L = \log(y/3)/2$. Forgetting either produces an incorrect inverse.

10.2 A resonance and an exact cancellation

For $f(x) = x + x^2 + x^3$, take gaps 1 and 2. At inverse power y^3 , the index $(2, 0)$ contributes 2 and $(0, 1)$ contributes -1 , leaving 1. At power y^4 , the index $(3, 0)$ contributes -5 and $(1, 1)$ contributes 5, so the entire block disappears. A power cutoff must be applied to weights, and terms of equal weight must be combined before output. A remainder bound based on a canceled boundary block is still valid, but is not sharp.

10.3 Several incommensurable gaps and logarithms

The model

$$f(x) = x + x^{\sqrt{2}}(1 + \log x) + 2x^{\sqrt{3}} \quad (10.2)$$

has gaps $\sqrt{2} - 1$ and $\sqrt{3} - 1$. The weights include each gap, twice the first gap, and their sum; logarithmic polynomials occur at some but not all weights. Formula (5.7) supplies the mixed correction. The package generates all weights below the requested cutoff without rationalizing either gap. An independent exact residual test for this example is included.

10.4 Other finite endpoints and inversion at infinity

For a right-sided branch at $x = x_0$, $y = y_0$, introduce $u = x - x_0 > 0$ and $v = y - y_0$. If $F(x_0 + u) - y_0$ has the required form, invert it and add x_0 . For a left-sided branch, choose $u = x_0 - x > 0$ and orient the target coordinate so that its leading coefficient is positive. These changes of variables are mathematical branch data; version 1.0 accepts the normalized endpoint zero and does not choose them automatically.

If a positive function grows to infinity and $\tilde{F}(t) = 1/F(1/t)$ has an admissible expansion at $t = 0^+$, then

$$F^{-1}(y) = \frac{1}{\tilde{F}^{-1}(1/y)}.$$

The forward reciprocal must first be expanded with its correct remainder, and inverse error must be transported through the final reciprocal. A normal expression alone does not retain that precision information.

11 Implementation and its exact contract

11.1 Accepted and rejected inputs

The main engine accepts a finite sum

$$f(x) = \sum_j x^{\alpha_j} P_j(\log x)$$

whose smallest nonzero block is ax^p , $a, p > 0$. All exponents and the target cutoff must be exact real algebraic constants. Logarithmic coefficients must be polynomials with nonnegative integer logarithmic degree; their coefficients may contain fixed real parameters when **Assumptions** proves reality and the leading coefficient's positivity. Inexact **Real** input is rejected rather than silently converted to exact rational data.

The parser intentionally rejects unexpanded analytic factors, symbolic exponents, negative powers of $\log x$, iterated logarithms, exponential sectors, and a leading logarithmic block. This is a supported-class boundary, not a statement that such functions have no inverse expansion. See Section 13 for the required extensions.

The source and target must be distinct unassigned symbols, and the forward expression must not contain the target symbol. User assumptions concern fixed parameters only, not either asymptotic variable. Positivity of source and target is already part of the selected branch. Use exact **Sqrt**[2], not a decimal approximation to it.

11.2 Public functions and data

Interface	Meaning
RealInverseAsymptotic [f , { x , 0}, { y , q }]	Construct the complete-block inverse expansion below the exclusive power cutoff q .
Normal [result]	The finite expression only; discard its remainder metadata.
result ["Remainder"]	Zero for an exact monomial inverse, otherwise an inert PowerLogRemainder descriptor.
result ["Terms"]	Pairs $\{\beta, C\}$ meaning $(y/a)^\beta C$, with logarithms already substituted as $\log(y/a)/p$.
result ["Branch"]	The positive branch with $x/(y/a)^{1/p} \rightarrow 1$.
InverseResidual [result]	Independent normalized sparse-composition residual below the result's relative cutoff.
InverseResidual [result , h]	The same calculation at a user-specified positive relative cutoff in the z coordinate. It can expose the first nonzero omitted residual.
PerturbativeInverse [h , { x }, { y , n }, eps]	Degree- n polynomial in a marker for $u + \mathbf{eps} \, h(u) = y$, with $u = y$ at marker zero.

The option **Assumptions** defaults to **True**, **InputRemainder** to **None**, and **MaxMultiIndices** to 20000. The last is a conservative resource budget for traversal, boundary candidates, and sparse products; it is not an exact count of final output blocks and not a global time limit. A budget failure returns **Failure**, never a result pretending to have reached the requested order.

Additional fields include "LeadingPower", "LeadingCoefficient", "InputInterpretation", "RelativeCutoff", "RemainderPower", "RemainderLogDegree", "MultiIndexCount", and "BoundaryCount". Internal scaled terms are retained for exact residual verification. The mathematical guarantee is conditional on the fixed-parameter hypotheses; it is neither an interval certificate nor a uniform parameter estimate.

11.3 The algorithm in six stages

First expand the finite input into monomial factors, replace $\log x$ temporarily by a fresh symbol, and gather equal powers. Prove reality of coefficient data and positivity of the constant leading block. Second, normalize by ax^p and compute the positive gaps. Third, enumerate I_H recursively with exact comparisons; prune a branch as soon as its weight reaches H . Fourth, evaluate (5.2) for every retained multi-index. Fifth, canonicalize and gather equal weights, cancel zero polynomials, and convert to the target variable. Sixth, compute the finite boundary and combine its error bound with any transported input uncertainty.

The core polynomial loop is small:

```
q = Expand[Times @@ MapThread[Power, {blocks, n}]];
Do[
  q = Expand[D[q, ell] + (1 + weight + p j) q],
  {j, 1, Total[n] - 1}
];
q = (-1)^Total[n] q /
  (p^Total[n] (Times @@ (Factorial /@ n)));
```

The surrounding validation and support arithmetic matter just as much as this formula. `RootReduce` is used to canonicalize exact algebraic weights, and comparisons must simplify to a definite truth value; there is no approximate tie-breaking rule [12]. The display layer uses `ToRadicals` when available, without changing the exact keys.

If $\delta_{\min} = \min_i \delta_i$, the maximum total degree retained is strictly smaller than $H/\delta_{\min}$. A coarse upper bound on the number of retained indices is

$$\prod_i (1 + \lfloor H/\delta_i \rfloor).$$

The actual simplex enumeration is smaller, but high dimension, tiny gaps, large logarithmic degrees, or complicated algebraic numbers can still make a request expensive. These are mathematical and representation costs, not evidence that a rational grid should be substituted for an irrational one.

11.4 Independent residual verification

For a computed finite U , the checker forms (6.1) without using the Lagrange coefficient formula. Sparse convolution is used for multiplication, and

$$(1 + U)^r = \sum_{j \geq 0} \binom{r}{j} U^j, \quad \log(1 + U) = \sum_{j \geq 1} \frac{(-1)^{j+1}}{j} U^j$$

are truncated as soon as their valuations exceed the cutoff. Polynomial substitution $B_i(L + \log(1 + U))$ is a finite Taylor expansion in its second summand. This checks normalization, logarithmic differentiation, mixed terms, and resonant cancellation by a different route.

The returned residual is

$$f(G_q(y))/y - 1.$$

If its relative cutoff is H , the corresponding cutoff for the unnormalized forward residual is $1 + H/p$. A zero residual below the cutoff confirms the finite formal identity for the supplied finite model. It does not prove the user's unspecified remainder bounds and does not assess floating-point rounding. If the result was constructed from a forward jet, the check is explicitly limited to that finite jet.

11.5 Why not manufacture a rational `SeriesData` object?

`SeriesData` stores powers on its documented index/denominator grid; logarithmic factors can appear in its coefficients [7]. That is useful for ordinary series and many logarithmic expansions, but it is not a faithful reason to round $1 + n(\sqrt{2} - 1)$ to rational exponents. The package uses its own transparent finite data object and explicit remainder descriptor instead. It also does not overload big-O arithmetic that it has not implemented.

12 Verification performed and reproducibility

12.1 What was actually executed

The connected Wolfram evaluator was attempted, but its MCP endpoint returned HTTP 404 before evaluating code. No local Wolfram kernel was available. Accordingly, the native `.wlt` suite is supplied but is *not* reported as executed, and there is no claim of a tested Mathematica version.

The independent Python verification script was executed. It uses SymPy for exact algebra and mpmath at 150 decimal digits for numerical comparisons. It reports 18 successful checks, including exact sparse-composition tests for the original logarithmic and irrational examples, mixed irrational gaps, resonances, two logarithmic blocks, a nonunit leading power, a fractional leading power, and a pure monomial. It independently compares the first six logarithmic corrections with the derivative form of Lagrange inversion, checks their signed Catalan leading coefficients, checks an ordinary quadratic inverse against its square-root formula, and verifies the first nonzero omitted logarithmic residual. The 24 numerical rows and 12 numerical checks of (8.7) are diagnostics, not certified interval calculations.

A lexical check of balanced delimiters, strings, and nested comments was also performed on the Wolfram sources. This detects a limited class of source defects; it does not execute Wolfram semantics or certify the parser. The limitations above are important even though independent mathematical checks are substantial. They are recorded in `verification/STATUS.md` and the machine-readable result file.

12.2 Numerical behavior on the positive branch

In Table 2, m is the largest retained ordinary power of y for the logarithmic example, and the scaled error is

$$\frac{|G_m(y) - g_1(y)|}{y^{m+1}(1 + |\log y|)^m}.$$

The reference inverse was obtained by high-precision bisection on the positive interval $[y/2, 2y]$, which brackets the root for the displayed arguments. Table 3 similarly uses N correction terms for the irrational example and divides the error by $y^{1+(N+1)(\sqrt{2}-1)}$. The tables illustrate convergence and the proven orders; they are not the proofs of those orders.

y	m	Absolute error	Error / stated scale
10^{-2}	2	2.4202×10^{-5}	0.770327
10^{-2}	3	1.8127×10^{-6}	1.02936
10^{-2}	5	1.2899×10^{-8}	2.33140
10^{-2}	7	1.0791×10^{-10}	6.20810
10^{-4}	2	1.2685×10^{-10}	1.21677
10^{-4}	3	2.4098×10^{-13}	2.26395
10^{-4}	5	1.1512×10^{-18}	10.3744
10^{-4}	7	6.6354×10^{-24}	57.3572
10^{-8}	2	5.8954×10^{-22}	1.56309
10^{-8}	3	2.4782×10^{-28}	3.38338
10^{-8}	5	5.8498×10^{-41}	21.1747
10^{-8}	7	1.6739×10^{-53}	160.652

Table 2: Positive-branch numerical diagnostics for the logarithmic example. Computations use 150 decimal digits; displayed values are rounded.

y	N	Absolute error	Error / stated scale
10^{-2}	1	2.5197×10^{-4}	1.14343
10^{-2}	2	5.9671×10^{-5}	1.82407
10^{-2}	3	1.5336×10^{-5}	3.15815
10^{-2}	4	4.1555×10^{-6}	5.76451
10^{-4}	1	6.6314×10^{-8}	1.36556
10^{-4}	2	2.3627×10^{-9}	2.20788
10^{-4}	3	9.0972×10^{-11}	3.85768
10^{-4}	4	3.6835×10^{-12}	7.08808
10^{-8}	1	3.3324×10^{-15}	1.41310
10^{-8}	2	2.6235×10^{-18}	2.29095
10^{-8}	3	2.2302×10^{-21}	4.01029
10^{-8}	4	1.9927×10^{-24}	7.37862

Table 3: Positive-branch numerical diagnostics for the irrational example. Computations use 150 decimal digits; displayed values are rounded.

12.3 Running the supplied checks

In a Wolfram environment, run

```
wolframscript -file Tests/run-tests.wls
```

or evaluate the equivalent `TestReport` call from a notebook. The supplied MUnit suite contains 36 tests covering expected coefficients, residuals, remainder metadata, symbolic real coefficients, precision caps, unsupported inputs, and resource-limit behavior. Results from this suite must be obtained in the user's own kernel; no pass count is asserted here.

For the independent checks,

```
python verification/verify.py --output verification
```

regenerates the JSON report and coefficient table. The script explicitly identifies itself as a reference implementation, not a Wolfram interpreter. The examples file contains the two original calls, a mixed model, a nonunit leading model, a forward-jet example, and a high-precision numerical comparison. The archive contains all LaTeX inputs needed to rebuild this PDF; it does not depend on repository macros or on redistributing fonts.

13 Boundaries of the theory implemented here

13.1 A small logarithmic correction with zero power gap

For $f(x) = x + x/\log x$, the correction is relatively small, but its power gap is zero. All corrections can occur at the same power y , with increasingly negative powers of $\log y$. The finite region I_H would no longer be finite in the power grading used here. The appropriate coordinate is $t = 1/\log y$, not another positive power of y . With $g = yW$,

$$W\left(1 + \frac{t}{1 + t \log W}\right) = 1,$$

which is analytic near $(W, t) = (1, 0)$ and can be reversed as a Taylor series in t . The main engine correctly rejects this input rather than attempting to enumerate infinitely many zero-weight contributions.

13.2 A leading logarithm, iterated logarithms, and flat sectors

A leading behavior such as $x^p(-\log x)^b$ requires a different dominant coordinate, often with a Lambert- W core or an implicit logarithmic normalization. The general identity (4.3) remains useful, but a theorem controlling its actual asymptotic grading must accompany it. Leading logarithms are not automatically reduced by version 1.0.

Iterated logarithms or rational functions of $\log x$ require a larger coefficient algebra and a second truncation policy. Exponentially small terms such as $e^{-1/x}$ require an exponential sector grading. They are not ordinary power-log coefficients, and silently dropping them would lose information beyond all algebraic orders. The marker utility can generate finite differential expressions for many such perturbations, but it does not assert that setting the marker equal to one has an established asymptotic remainder. This is the precise distinction between a formula generator and a transseries engine with a proved support contract.

13.3 Symbolic exponents and nonuniform limits

When an exponent depends on a parameter, the ordering of support values can change at resonances, and a positive gap may approach zero. A symbolic exponent engine would need to partition parameter space into cells with proved ordering relations, handle boundary cells separately, and carry uniformity conditions into error bounds. The present exact algebraic restriction avoids making unsupported decisions about such orderings. The mathematics of the master formula still applies to any fixed positive real gaps, but that does not mean a symbolic implementation can compare all parameter-dependent weights without additional information.

13.4 What a larger implementation should add next

The most direct extension is a sparse Newton solver for the same supported class, using the already implemented composition operations and a unit reciprocal routine. It should be benchmarked against the explicit formula on high-order and many-gap examples. A second extension is structured forward-jet arithmetic, so elementary functions can be expanded together with their differentiated remainder contracts rather than relying on a manual `InputRemainder` declaration. Larger logarithmic and exponential coefficient algebras should come with their own exact support comparators and error semantics, not merely additional parser patterns. Finally, interval evaluation of the residual and a proved derivative bound would turn the a posteriori theory into computable numerical certificates. None of these additions is falsely advertised as present in version 1.0.

14 Relationship to the supplied materials and main conclusions

The supplied repository directory contains a canonical transseries volume and a combinatorial companion. The group README was read, and the relevant phase-coordinate, multiplicative-sector, and scaled-reversion portions of the companion source were inspected [5, 6]. In particular, the companion’s labels `t1:thm:phase`, `t1:thm:multi`, and `t2:prop:scaled` explain the operator and marker methodology. The README explicitly warns against conflating formal identities, analytic convergence, and the scope of formalized results.

The multi-megabyte canonical source could not be read through the available repository-file endpoint; this report does not claim a complete review of that volume. It instead gives self-contained proofs of every specialized inversion and remainder result it uses. No new Lean formalization is claimed, and the kernel checks of this Wolfram package are separate from any formalization claims made by the repository.

The classical Lagrange identity is not new. The contribution here is its explicit positive-endpoint specialization with arbitrary positive gaps, nonunit monomial cores, a directly enumerable exact support, complete logarithmic blocks, a finite-boundary remainder law, guarded forward precision, and an auditable package interface tailored to the question. The analysis separates three assertions that must never be confused: coefficient correctness, realization as an asymptotic expansion of a specified real branch, and successful execution in a particular CAS. The first two are proved for the stated class and checked independently; the third still requires the supplied suite to be run in an actual Wolfram kernel.

The practical answer is consequently not a request to persuade `InverseFunction` to guess the intended branch. Select the positive leading root, compute finite power–log blocks in their natural support, retain the logarithms in the remainder, and verify the forward residual. Both original examples, and finite mixtures substantially more complicated than either one, fit this construction.

A A compact marker recipe

For the specific near-identity class $f(x) = x + h(x)$, the following minimal recipe is useful even independently of the full package:

```
ClearAll[nearIdentityPolynomial];
nearIdentityPolynomial[h_, x_Symbol, y_Symbol, n_Integer] /; n >= 0 :=
  y + Total[Table[
    (-1)^j D[(h /. x -> y)^j, {y, j - 1}]/j!,
    {j, 1, n}
  ]];

nearIdentityPolynomial[x^2 (1 + Log[x]), x, y, 4] // Expand
```

The table uses numerical integer loop indices before taking the derivatives. For the original logarithmic perturbation, the result has the remainder in (8.3); for a general h , this code alone does not establish an asymptotic grading. The package's `PerturbativeInverse` keeps an explicit marker to make that qualification visible.

B An exact residual bracket lemma

Lemma B.1 (Existence and an error enclosure from a residual). *Let F be continuously differentiable on $[A, B]$, with $F' \geq m > 0$ there. Let $x_0 \in (A, B)$ and $r = F(x_0) - y$. If*

$$[x_0 - |r|/m, x_0 + |r|/m] \subseteq [A, B],$$

then $F(x) = y$ has a unique solution in that interval, and $|x - x_0| \leq |r|/m$.

Proof. Integrating the derivative lower bound gives $F(x_0 + |r|/m) \geq F(x_0) + |r| \geq y$ and $F(x_0 - |r|/m) \leq F(x_0) - |r| \leq y$. The intermediate value theorem supplies a root; strict monotonicity gives uniqueness. The stated interval is the error enclosure. $\square$

This lemma establishes existence as well as an error bound, unlike a mean-value estimate that presupposes a nearby root. For the first example global existence has already been proved, so (8.7) is available directly. For general models, an explicit interval and derivative bound would be extra computable inputs to a numerical certification layer.

C Source and verification manifest

Archive member	Purpose
Kernel/RealInverseAsymptotics.wl	Main package and independent formal residual arithmetic.
Kernel/init.m	Conventional package loader.
Examples/examples.wl	Reproducible Wolfram calls and numerical diagnostic.
Tests/RealInverseAsymptotics.wlt	36 native MUnit regression tests; not executed in this session.
Tests/run-tests.wls	Native test runner with an exit status.
verification/verify.py	Independently executed exact and numerical reference checks.
verification/verification_results.json	Recorded Python results, versions, coefficients, and numerical values.

Archive member	Purpose
verification/STATUS.md	Explicit boundary between executed checks and unexecuted native tests.
article/real_inverse_asymptotics.tex	Main manuscript source.
article/numeric_log.tex, numeric_irr.tex	Generated table inputs used by the manuscript.
article/real_inverse_asymptotics.pdf	Compiled, visually inspected report.
README.md, LICENSE	Usage, build instructions, scope, and licensing.

References

- [1] V. Reshetnikov, *How to get an asymptotic of the real-valued branch of the inverse function?* Mathematica Stack Exchange, question 236367, 11 December 2020, updated 13 December 2020. User-supplied archive and public page consulted 7 September 2026. <https://mathematica.stackexchange.com/questions/236367/>.
- [2] I. M. Gessel, *Lagrange inversion*, Journal of Combinatorial Theory, Series A **144** (2016), 212–249. <https://arxiv.org/abs/1609.05988>.
- [3] G. A. Edgar, *Transseries for beginners*, arXiv:0801.4877, 2008, revised 2009. Expository background on ordered transseries, differentiation, and composition. <https://arxiv.org/abs/0801.4877>.
- [4] G. A. Edgar, *Transseries: Composition, Recursion, and Convergence*, arXiv:0909.1259, 2009. Background on compositional inversion and its formal setting; the specialized proofs in this report are independent. <https://arxiv.org/abs/0909.1259>.
- [5] V. Reshetnikov / ProveIt project, *Series and transseries*, group README, consulted 7 September 2026; retrieved blob 516aaa30e84cb0036a00f803cb6e8e3026febb8e. <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/series-and-transseries>.
- [6] V. Reshetnikov / ProveIt formal-development synthesis, *Combinatorial Transseries and Their Inverses*, consolidated companion volume, 4 September 2026. Relevant source sections on phase-coordinate inversion and scaled local reversion; retrieved source blob 05c56ce5ee60788a67f365110b62810550cba817. https://github.com/VladimirReshetnikov/ProveIt/blob/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/series-and-transseries/Combinatorial_Transseries_Inverses/Combinatorial_Transseries_Inverses.tex.
- [7] Wolfram Research, *SeriesData*, Wolfram Language documentation, consulted 7 September 2026. <https://reference.wolfram.com/language/ref/SeriesData.html>.
- [8] Wolfram Research, *Series*, Wolfram Language documentation, consulted 7 September 2026. <https://reference.wolfram.com/language/ref/Series.html>.
- [9] Wolfram Research, *InverseSeries*, Wolfram Language documentation, consulted 7 September 2026. <https://reference.wolfram.com/language/ref/InverseSeries.html>.
- [10] Wolfram Research, *InverseFunction*, Wolfram Language documentation, consulted 7 September 2026. <https://reference.wolfram.com/language/ref/InverseFunction.html>.
- [11] Wolfram Research, *Power*, Wolfram Language documentation, consulted 7 September 2026. <https://reference.wolfram.com/language/ref/Power.html>.
- [12] Wolfram Research, *RootReduce* and *Algebraics*, Wolfram Language documentation, consulted 7 September 2026. <https://reference.wolfram.com/language/ref/RootReduce.html>; <https://reference.wolfram.com/language/ref/Algebraics.html>.